

Gravity as Relaxation: Deriving General Relativity via Vacuum Mechanics

python 3.10+

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Can the Schwarzschild metric, gravitational light deflection, black hole photon rings, and Mercury's perihelion precession be derived without tensor calculus, purely by relaxing a physical vacuum medium on a discrete grid?

This repository provides an open-source, educational numerical framework demonstrating that the classical observational tests of General Relativity (GR) emerge naturally as the steady-state equilibrium of a physical vacuum medium undergoing classical relaxation transport.

Executive Overview

In standard gravitational pedagogy, calculating relativistic effects—such as the $1.75''$ Eddington solar light deflection or Mercury's $43''/\text{century}$ perihelion shift—requires Riemannian differential geometry, Christoffel symbols, and non-linear tensor field equations.

Gravity as Relaxation bridges the gap between everyday physics and relativistic astrophysics:

- Steady-State Equivalence:** Outside a spherical mass, Laplace's equation for gravity ($\nabla^2\Phi = 0$) is mathematically identical to steady-state thermal conduction ($\nabla^2T = 0$).
- Compactified Coordinates ($u = 1/r$):** The 3D spherical metric expansion factor r^2 cancels out identically under the coordinate inversion $u = 1/r$, mapping an infinite 3D space ($r \in [r_0, \infty)$) onto a finite 1D computational rod ($u \in [0, 1/r_0]$) with natural adaptive mesh refinement.
- Non-Linear Vacuum Stiffening:** In strong gravity, linear transport predicts an unphysical negative speed of light at the horizon ($c_{\text{eff}} \rightarrow -3.0 c_0$). Introducing a physical conductance factor $\sigma(\eta) = 1/\sqrt{\eta}$ (where η is the space-quantum packing density) self-regulates the medium, producing a smooth event horizon at $\eta = 4.0$ and $c_{\text{eff}} = 0.0$ with zero singularities.

4. **Acceleration as Vacuum Gradient:** Gravitational acceleration is shown to be the spatial slope of the coordinate speed of light:

$$\vec{a} = -c_{\text{eff}} \nabla c_{\text{eff}} = -\frac{1}{2} \nabla (c_{\text{eff}}^2) \approx +c_0^2 \nabla \eta$$

5. **Zero-Dependency Architecture:** All numerical algorithms—including the non-linear Red-Black SOR solvers, the 4th-order Runge-Kutta (RK4) ray-tracers, and orbital integrators—are implemented in **pure NumPy and Matplotlib**, requiring zero external C-compilers or heavy libraries (such as SciPy).

Repository Architecture

```
gravity-as-relaxation/
├── 01_thermal_rod/
│   └── simulate_thermal_rod.py           # Step 1: 1D linear heat relaxation (Gauss-Seidel)
├── 02_thermal_sphere/
│   ├── simulate_uniform_sphere.py      # Step 2A: 3D spherical shell on uniform radius
│   └── simulate_inverted_sphere.py     # Step 2B: Exact mapping of 3D sphere to 1D rod
├── 03_solar_weak_field/
│   ├── simulate_solar_medium.py        # Step 3A: Sun as a refractive medium (1.27 km/s)
│   └── simulate_earth_gravity.py       # Step 3B: Earth gravity profile & acceleration
├── 04_vacuum_density_bh/
│   ├── simulate_1field_relaxation.py    # Step 4A: Non-linear vacuum stiffening (sigma)
│   └── visualize_split_steps.py        # Step 4B: 4-step comparative breakdown (Numerical)
├── 05_ray_tracing/
│   ├── trace_sun_deflection.py          # Step 5A: 1.75'' Eddington solar light deflection
│   └── trace_black_hole_rings.py       # Step 5B: Strong-field black hole optics & photon paths
├── 06_mercury_precession/
│   └── simulate_mercury_orbit.py        # Step 6: Mercury's rosette orbit & 43''/century
├── images/                             # High-resolution simulation figures generated
├── README.md
└── requirements.txt
```

The 6 Progressive Steps

Step 1: The 1D Steady-State Thermal Rod

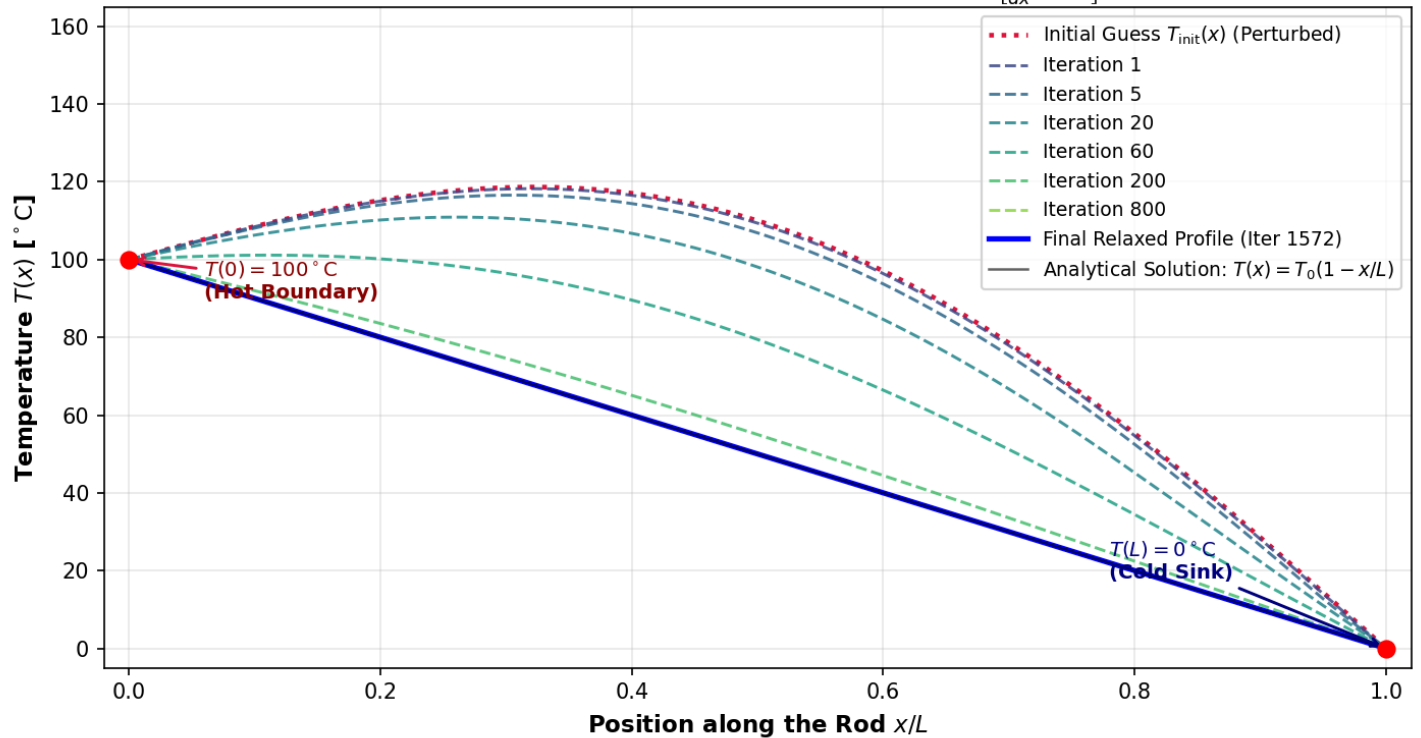
Script: 01_thermal_rod/simulate_thermal_rod.py

Theory: Steady-state heat conduction along a uniform rod:

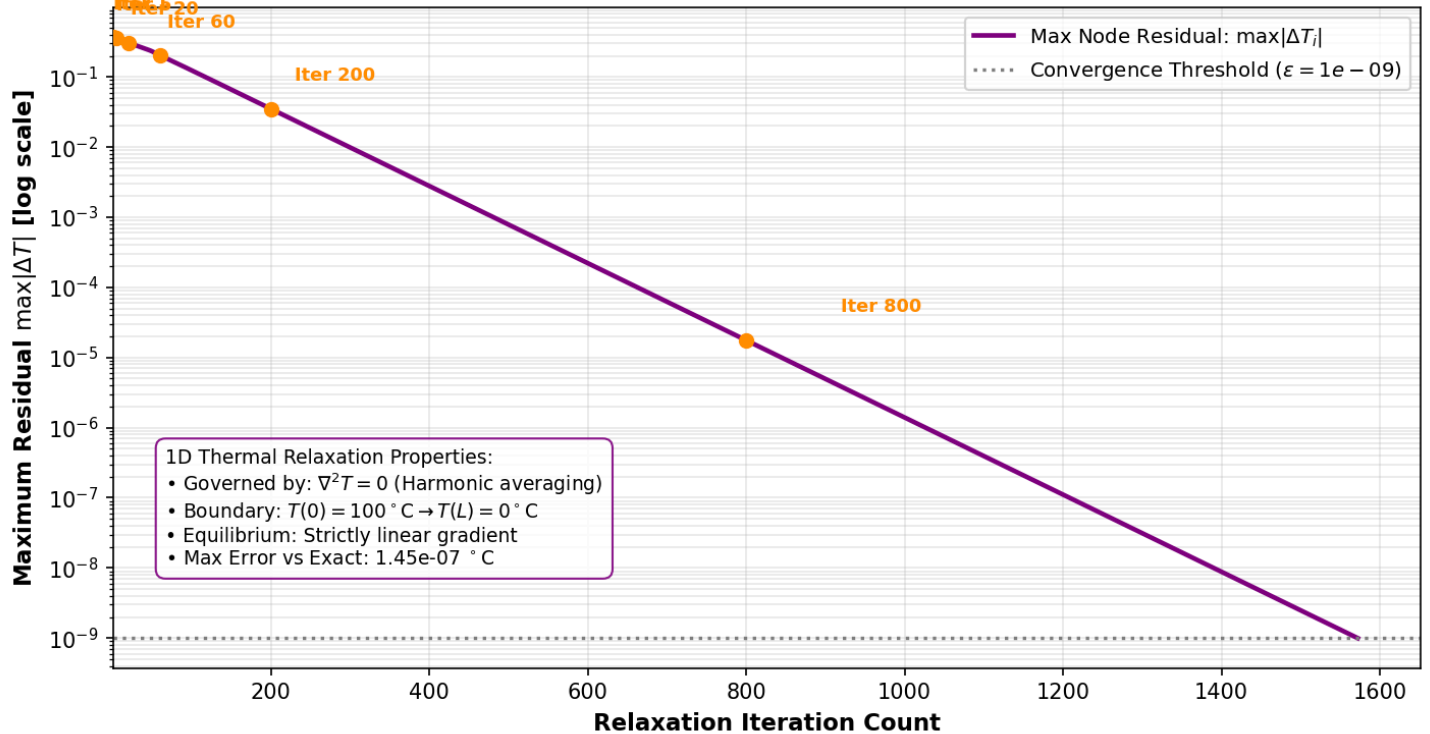
$$\frac{d^2T}{dx^2} = 0$$

- **Boundary Conditions:** Hot reservoir $T(0) = 100^\circ\text{C}$; Cold sink $T(L) = 0^\circ\text{C}$.
- **Mechanism:** Iterative Gauss-Seidel / Successive Over-Relaxation (SOR) smooths an arbitrary initial perturbation down to machine precision ($\epsilon < 10^{-9}$), proving that a strictly linear gradient $T(x) = T_0(1 - x/L)$ is the global equilibrium attractor of the Laplace operator.

Step 1: Relaxation of 1D Heat Equation $\left[\frac{d^2T}{dx^2} = 0\right]$



Numerical Convergence to Steady-State Equilibrium



Step 2: The Spherical Cavity & The $1/r$ Inversion

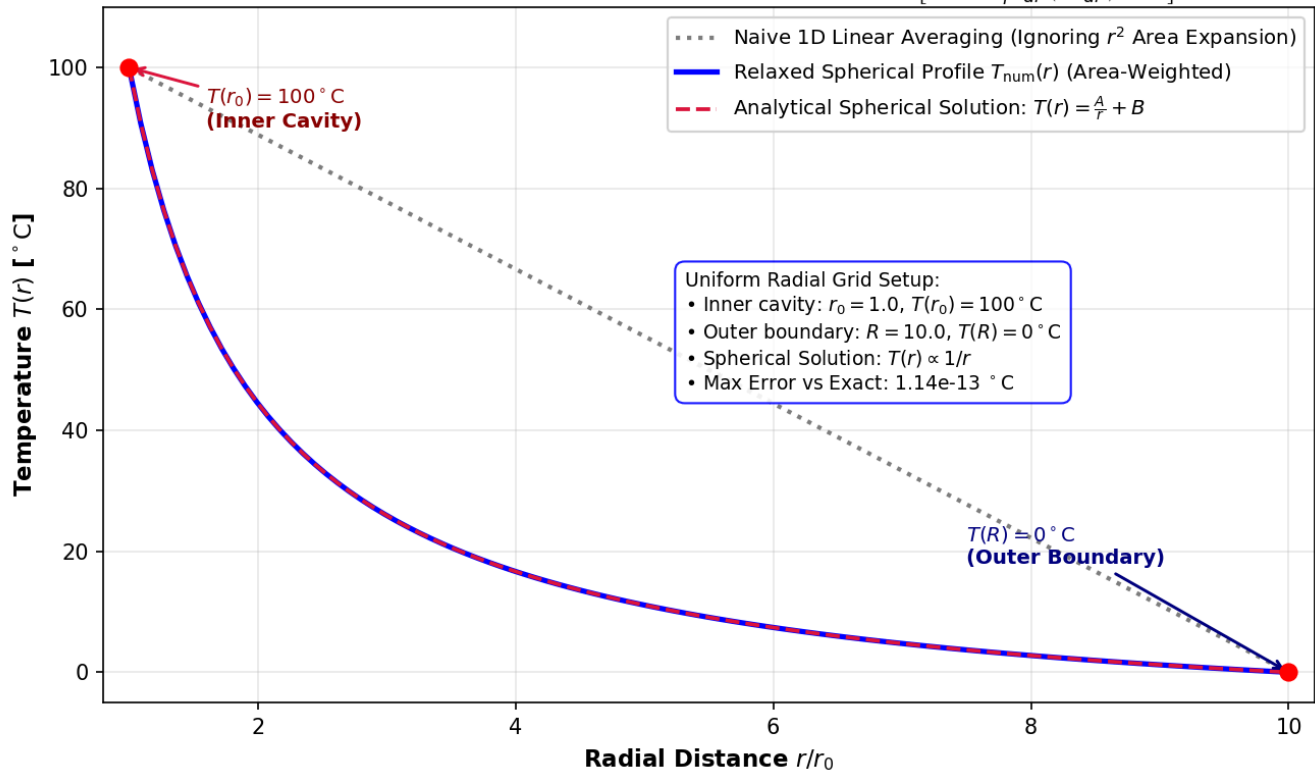
Directory: 02_thermal_sphere/

Theory: Radial diffusion through a 3D spherical shell:

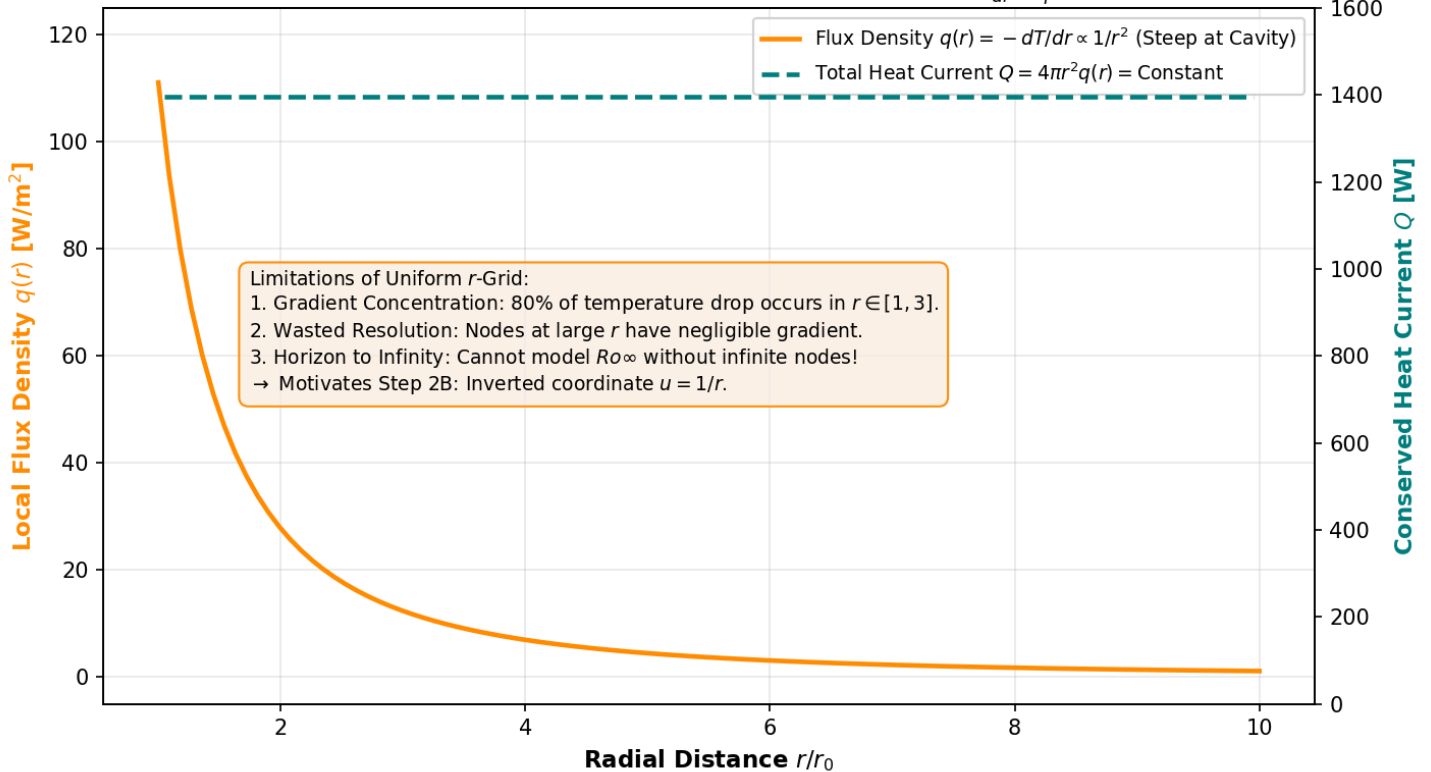
$$\nabla^2 T = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0$$

- **Part 2A (`simulate_uniform_sphere.py`):** On a uniform radial grid, heat current conservation ($Q = 4\pi r^2 q = \text{const}$) forces local flux density to dilute as $q(r) \propto 1/r^2$. Over 80% of the gradient clusters immediately near the cavity ($r \in [1, 3]$), wasting resolution in the outer regions and preventing the boundary from reaching infinity ($R \rightarrow \infty$).

Step 2A: Heat Conduction in a Spherical Shell $[\nabla^2 T = \frac{1}{r^2} \frac{d}{dr} (r^2 \frac{dT}{dr}) = 0]$



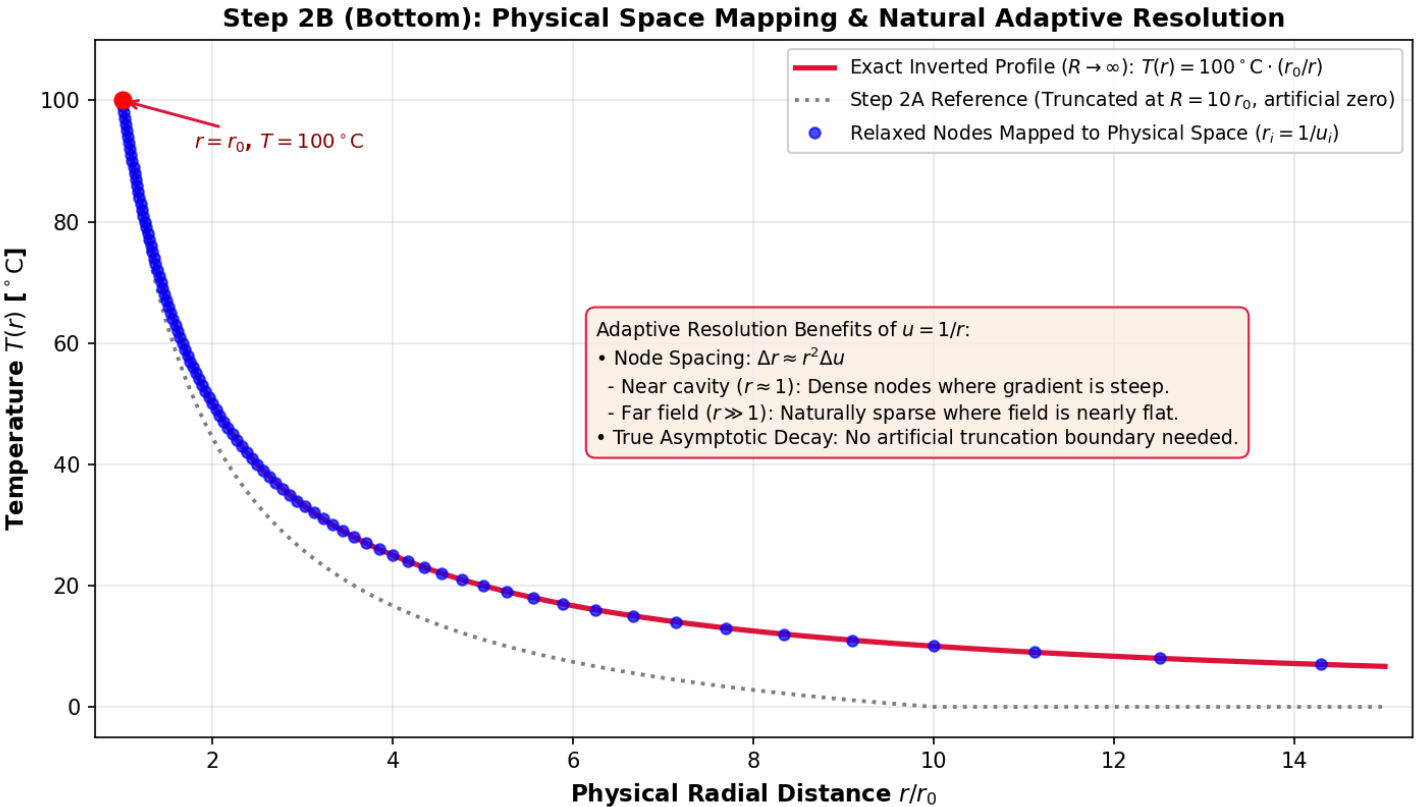
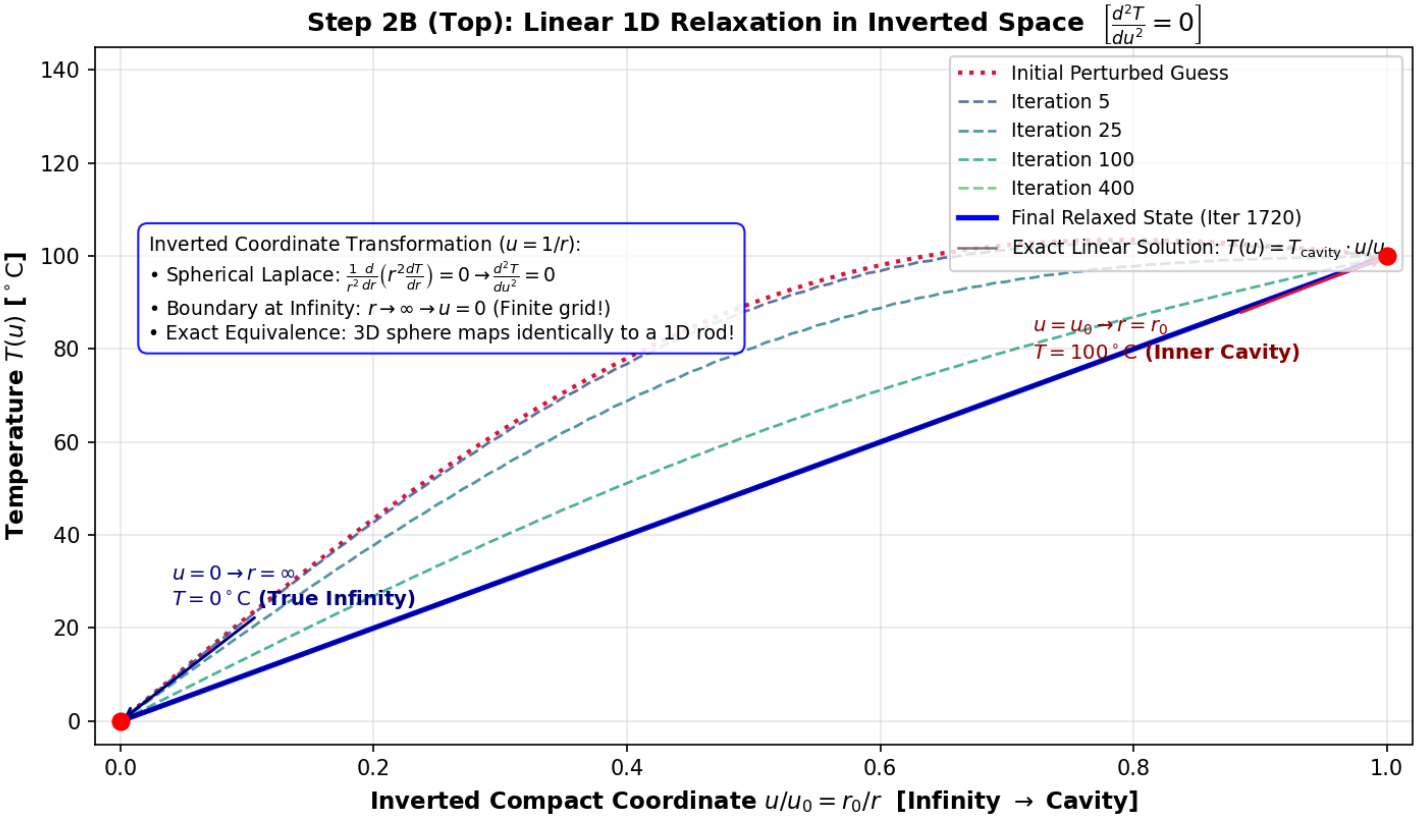
Physical Mechanism: Spherical Area Dilution Forces $\frac{dT}{dr} \propto \frac{1}{r^2} \rightarrow T(r) \propto \frac{1}{r}$



- **Part 2B (`simulate_inverted_sphere.py`):** Applying the coordinate transformation $u = 1/r$ collapses the 3D spherical problem into an exact 1D rod:

$$\frac{du}{dr} = -u^2 \implies r^2 \frac{dT}{dr} = -\frac{dT}{du} \implies \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = u^2 \frac{d^2 T}{du^2} = 0 \implies \frac{d^2 T}{du^2} = 0$$

The infinite domain ($r \in [r_0, \infty)$) is mapped to a compact interval ($u \in [0, 1/r_0]$), naturally concentrating nodes near the inner boundary where gradients are steepest ($\Delta r \approx r^2 \Delta u$).



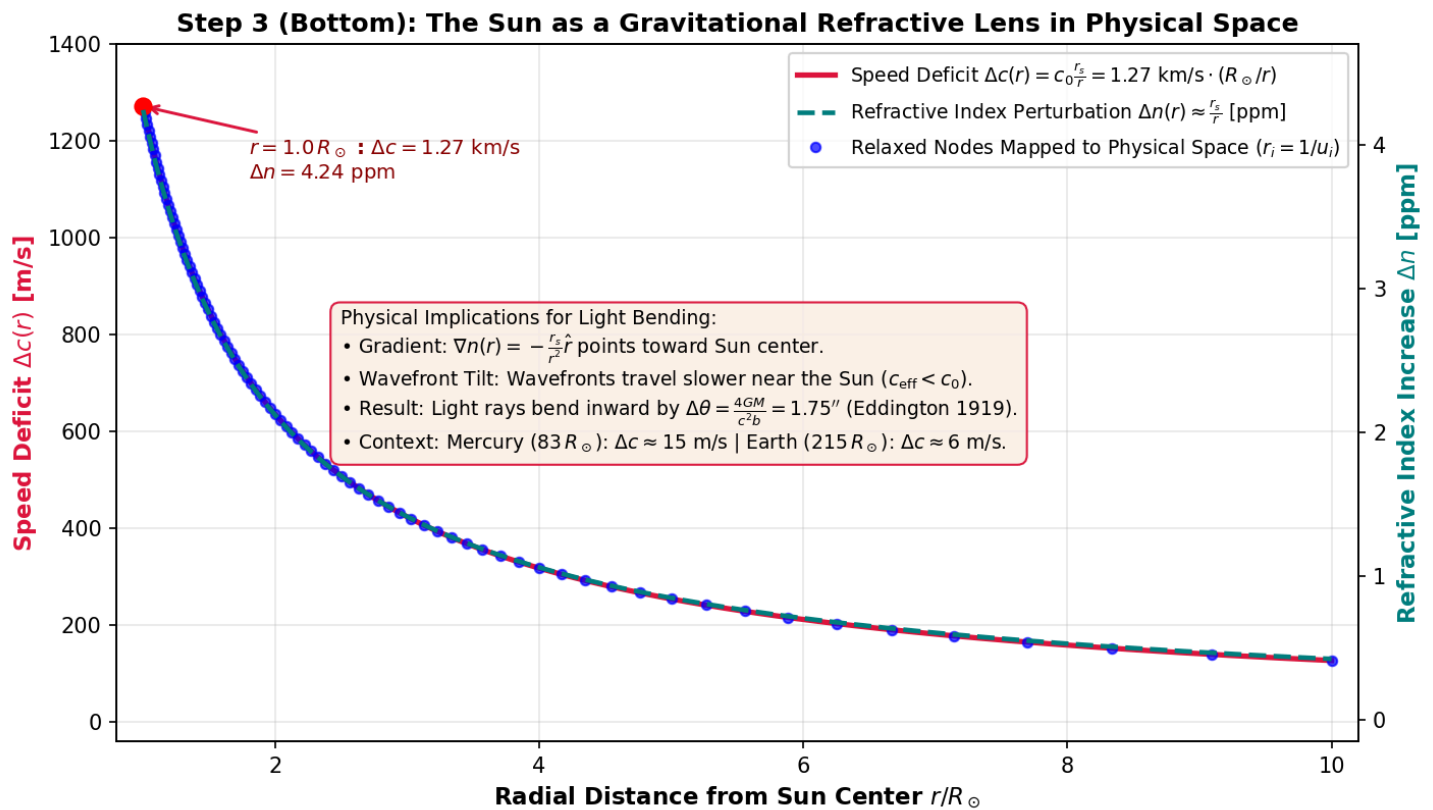
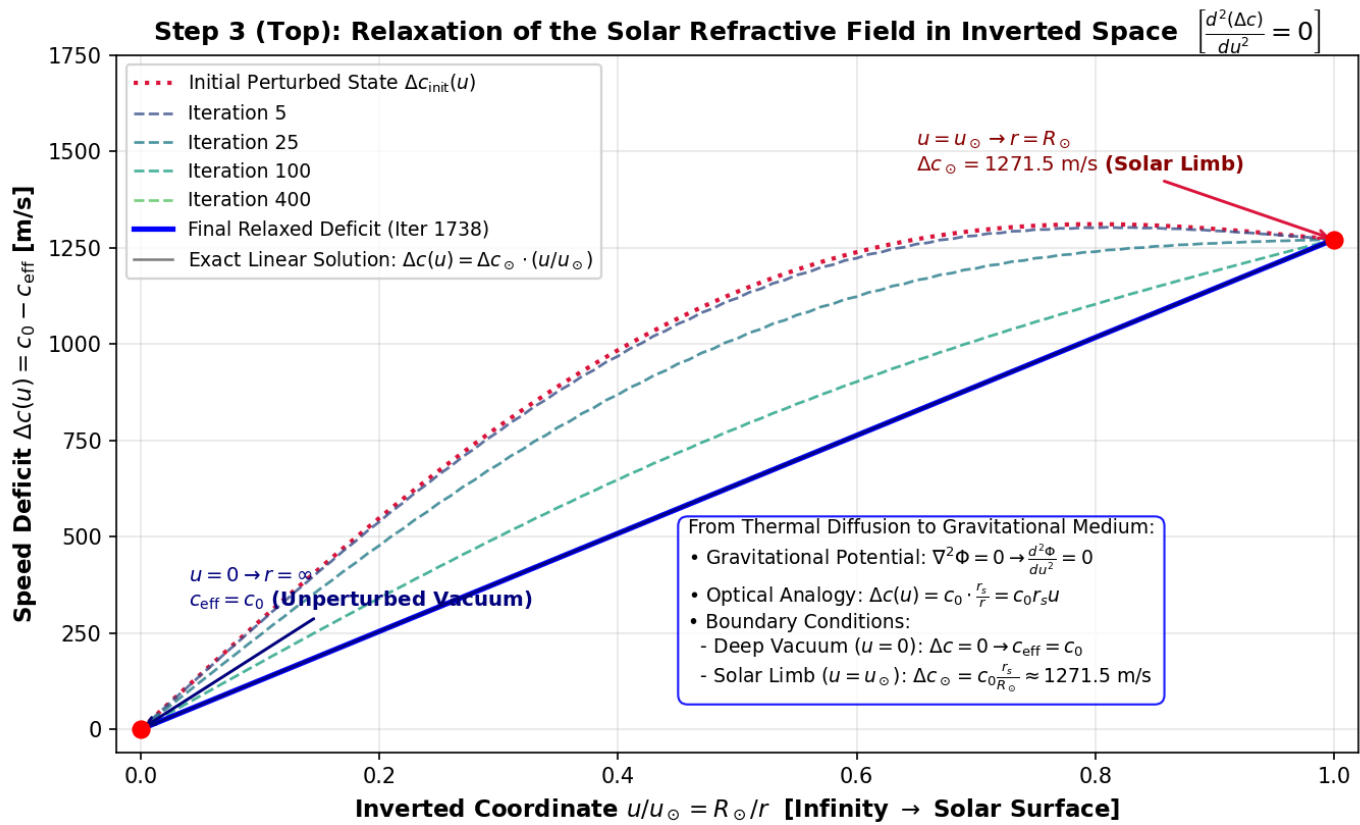
Step 3: From Heat to Weak Gravity (The Sun & The Earth)

Directory: 03_solar_weak_field/

Theory: Gravitational potential outside a spherical mass satisfies Laplace's equation: $\nabla^2 \Phi = 0 \implies \frac{d^2 \Phi}{du^2} = 0$. In Einstein's weak-field optical analogy:

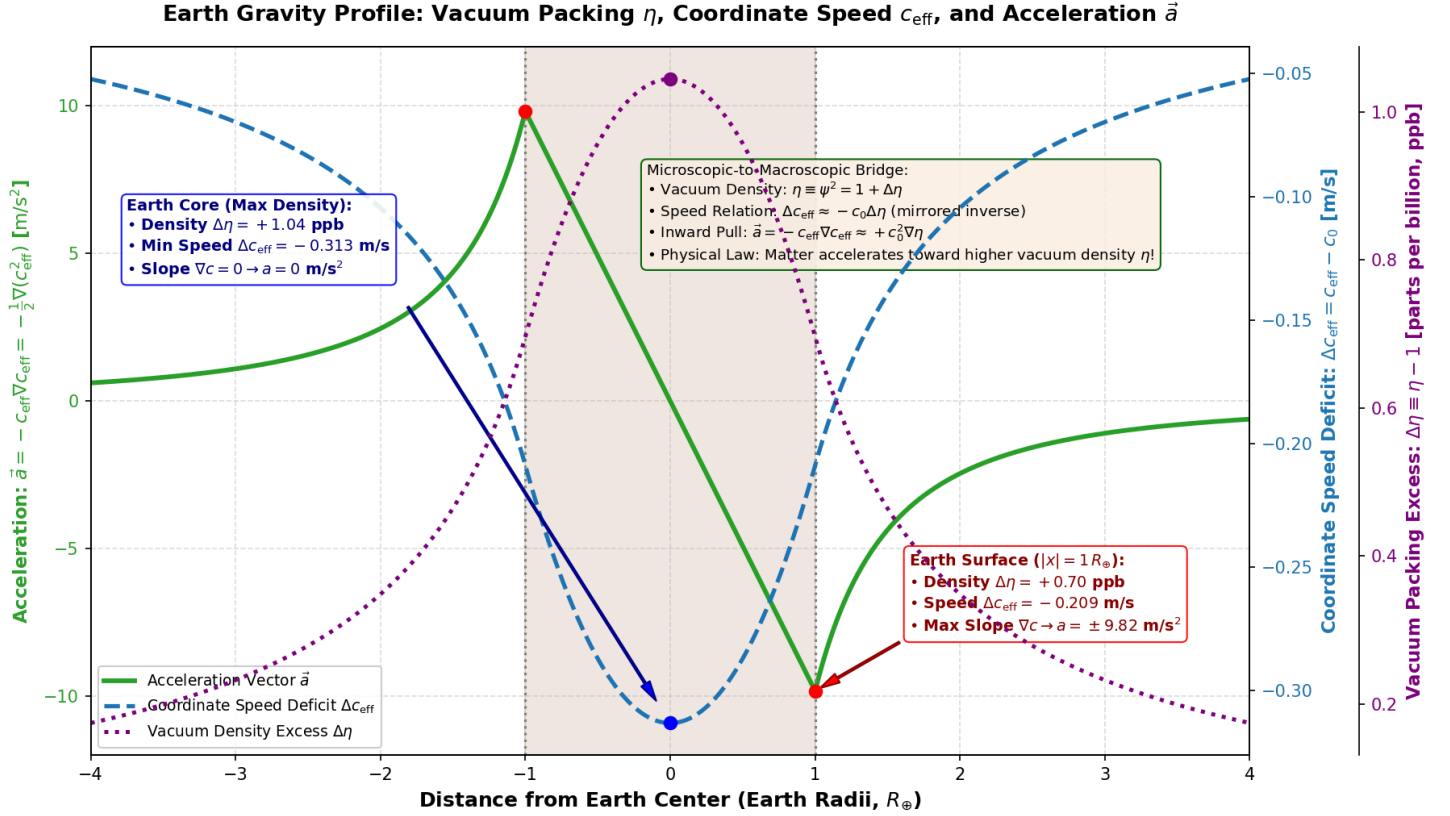
$$c_{\text{eff}}(r) \approx c_0 \left(1 - \frac{r_s}{r}\right) = c_0(1 - r_s u)$$

- **Part 3A (`simulate_solar_medium.py`):** The speed-of-light deficit $\Delta c(u) = c_0 - c_{\text{eff}}(u)$ relaxes linearly on the inverted rod. Light slows down by 1.27 km/s ($\Delta n \approx 4.24$ ppm) at the solar limb, producing an inward gradient that tilts electromagnetic wavefronts toward the mass center.



- **Part 3B (`simulate_earth_gravity.py`):** Connects the coordinate speed of light directly to everyday surface gravity on Earth ($g = 9.82 \text{ m/s}^2$). Evaluates the continuous optical landscape through the interior and exterior of Earth, verifying that acceleration is the spatial derivative of the coordinate speed of light:

$$\vec{a} = -c_{\text{eff}} \nabla c_{\text{eff}} = -\frac{1}{2} \nabla (c_{\text{eff}}^2) \approx +c_0^2 \nabla \eta$$



Step 4: Strong Gravity & Non-Linear Vacuum Stiffening

Directory: 04_vacuum_density_bh/

Theory: In strong fields, linear extrapolation $c_{\text{eff}}(s) = c_0(1 - r_s s)$ plunges to an unphysical negative value ($c_{\text{eff}} \rightarrow -3.0 c_0$) at the horizon ($s_h = 4/r_s$).

The vacuum medium **self-stiffens**: space-quantum packing density η reduces conductance as $\sigma(\eta) = 1/\sqrt{\eta}$, giving the non-linear flux balance:

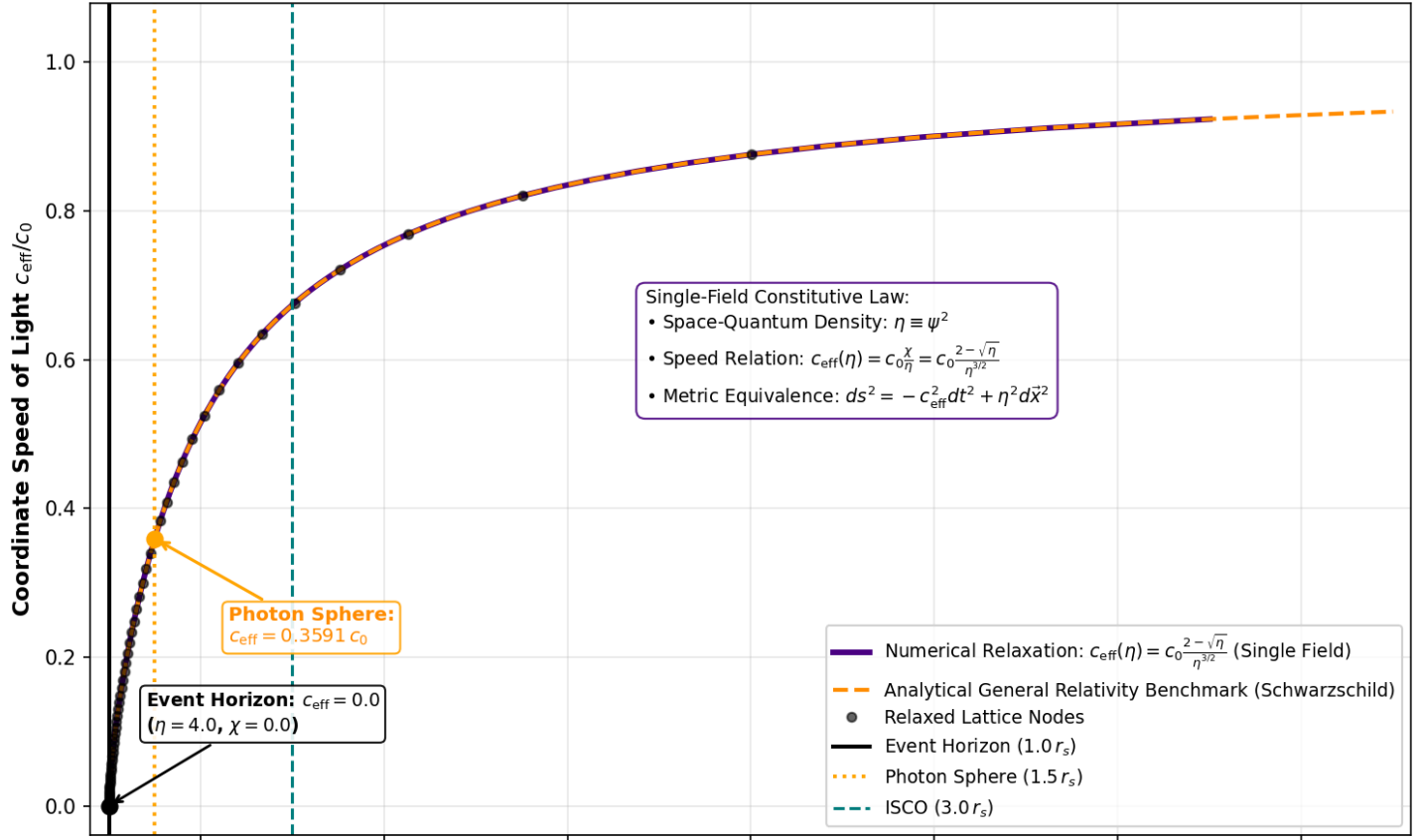
$$\frac{d}{ds} \left[\frac{1}{\sqrt{\eta}} \frac{d\eta}{ds} \right] = 0 \quad \text{on } s = 1/\rho \in [0, 4/r_s]$$

• Single-Field Decomposition:

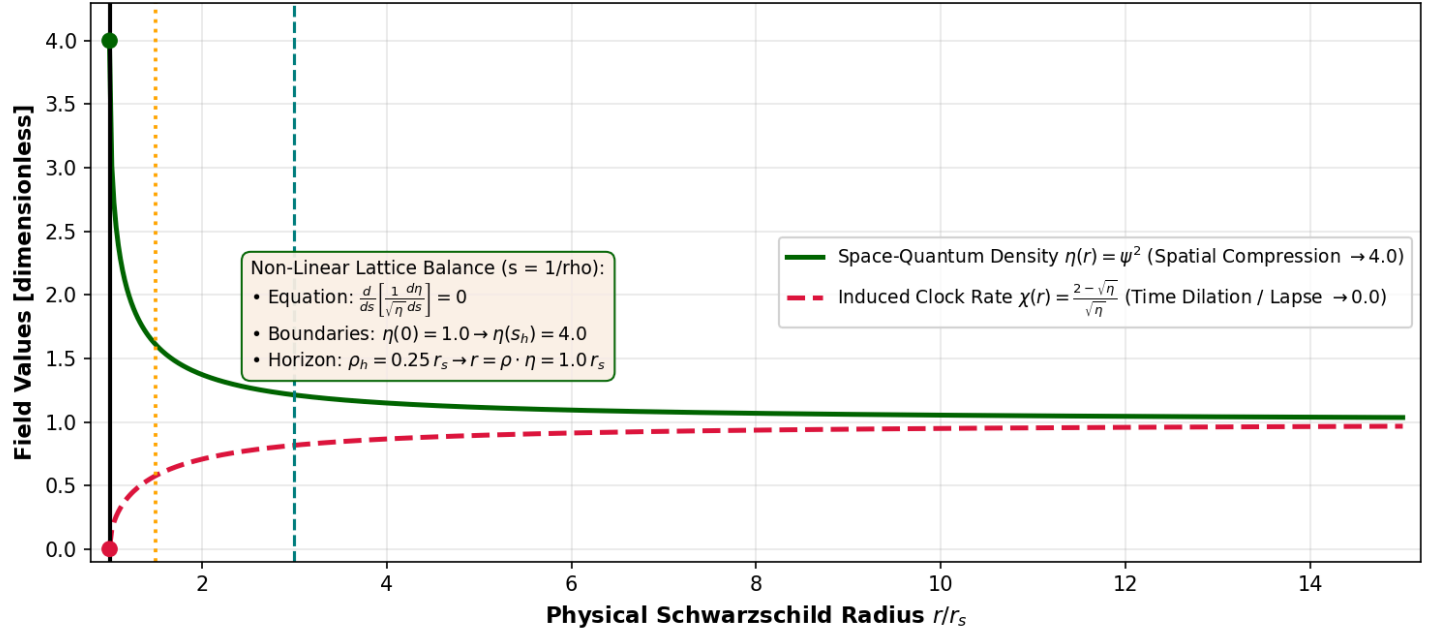
- Packing density: $\eta(r) = \psi^2 \rightarrow 4.0$ at the horizon.
- Clock lapse rate: $\chi(r) = \frac{2-\sqrt{\eta}}{\sqrt{\eta}} \rightarrow 0.0$ at the horizon (time freezes).
- Emergent coordinate light speed:

$$c_{\text{eff}}(\eta) = c_0 \frac{\chi}{\eta} = c_0 \frac{2 - \sqrt{\eta}}{\eta^{3/2}}$$

Step 4: Emergence of the Schwarzschild Coordinate Speed from Vacuum Relaxation

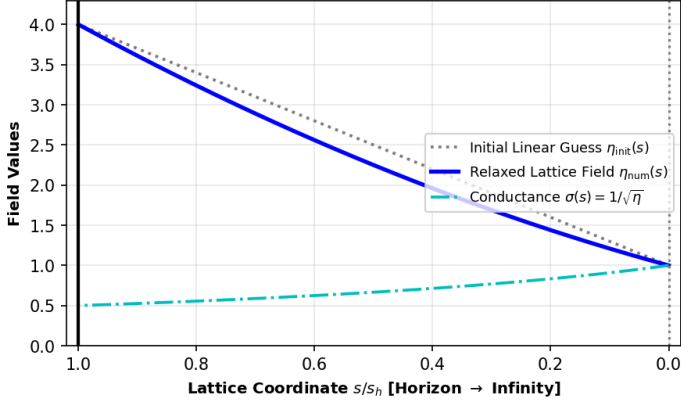


Microscopic Dual-Mechanism in Physical Space: Space Compression $\eta(r)$ vs Clock Lapse $\chi(r)$

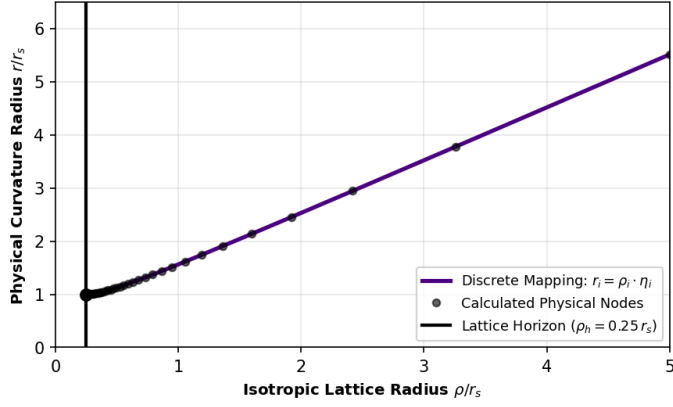


LEFT COLUMN: NUMERICAL METHOD (Vacuum Lattice Relaxation)

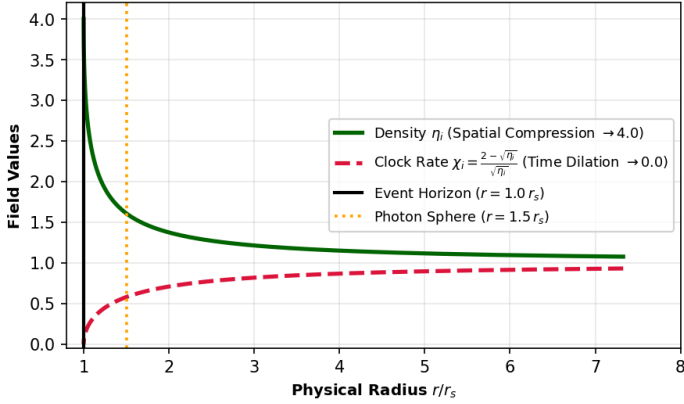
Step 1: Non-linear Flux Relaxation in $s = 1/\rho$



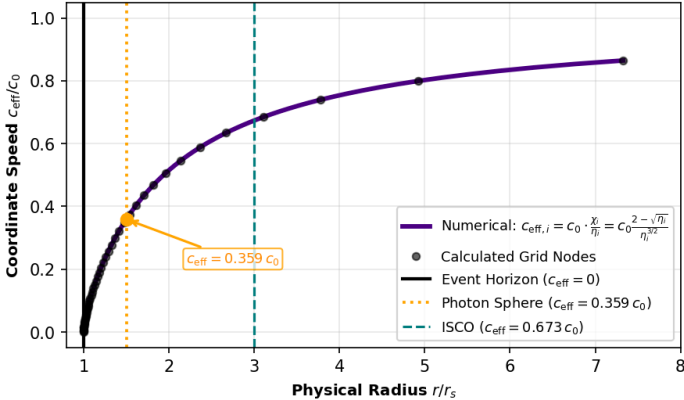
Step 2: Lattice Mapping to Physical Space



Step 3: Numerical Decomposition (Space η vs. Time χ)

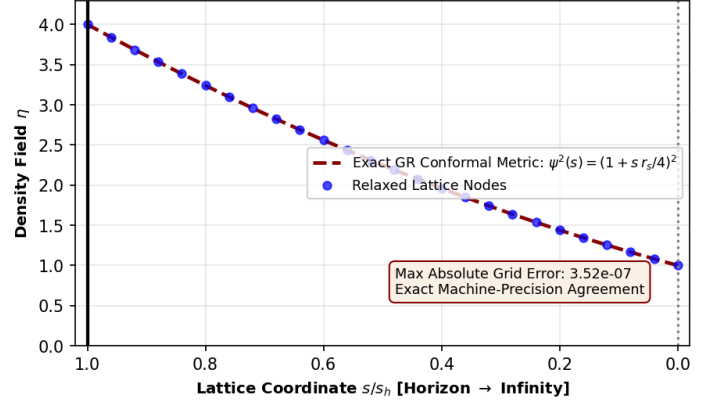


Step 4: Emergent Coordinate Light Speed $c_{eff}(r)$

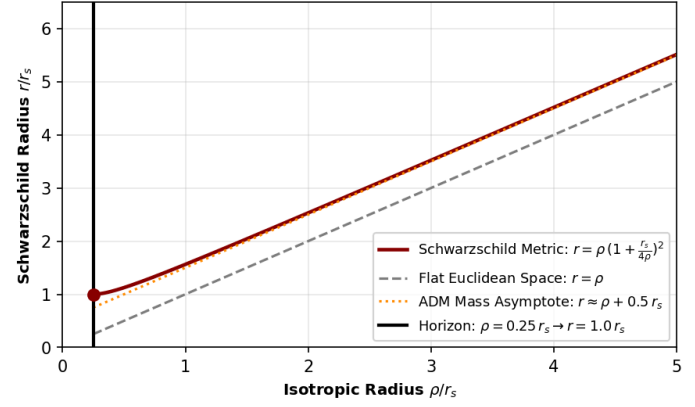


RIGHT COLUMN: THEORETICAL BENCHMARK (General Relativity / Schwarzschild)

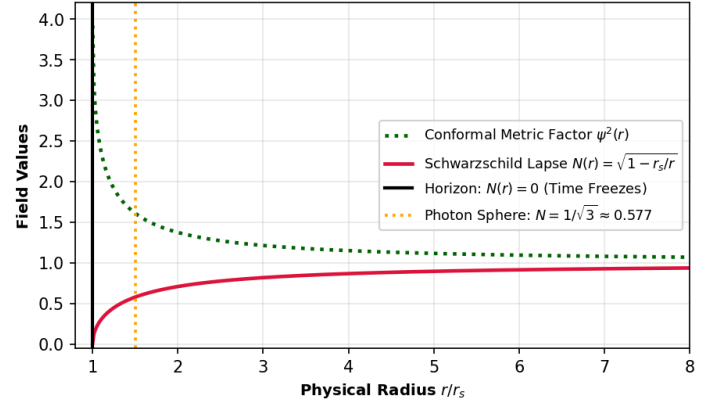
Validation 1: Equivalence to Conformal Scale ψ^2



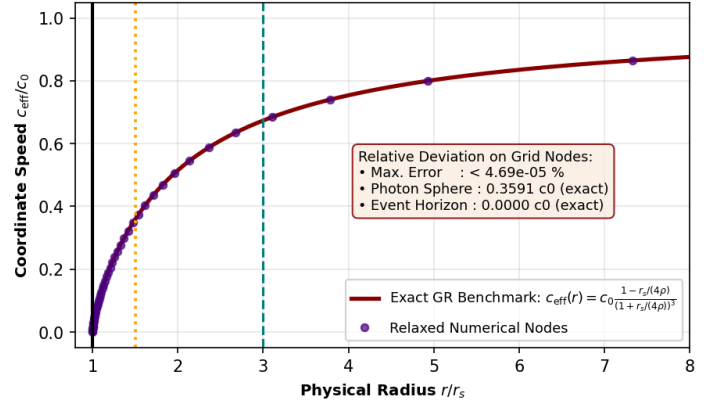
Validation 2: Schwarzschild Curvature & ADM Shift



Validation 3: Equivalence to Einstein Lapse $N(r)$



Validation 4: Verification Against Schwarzschild Metric



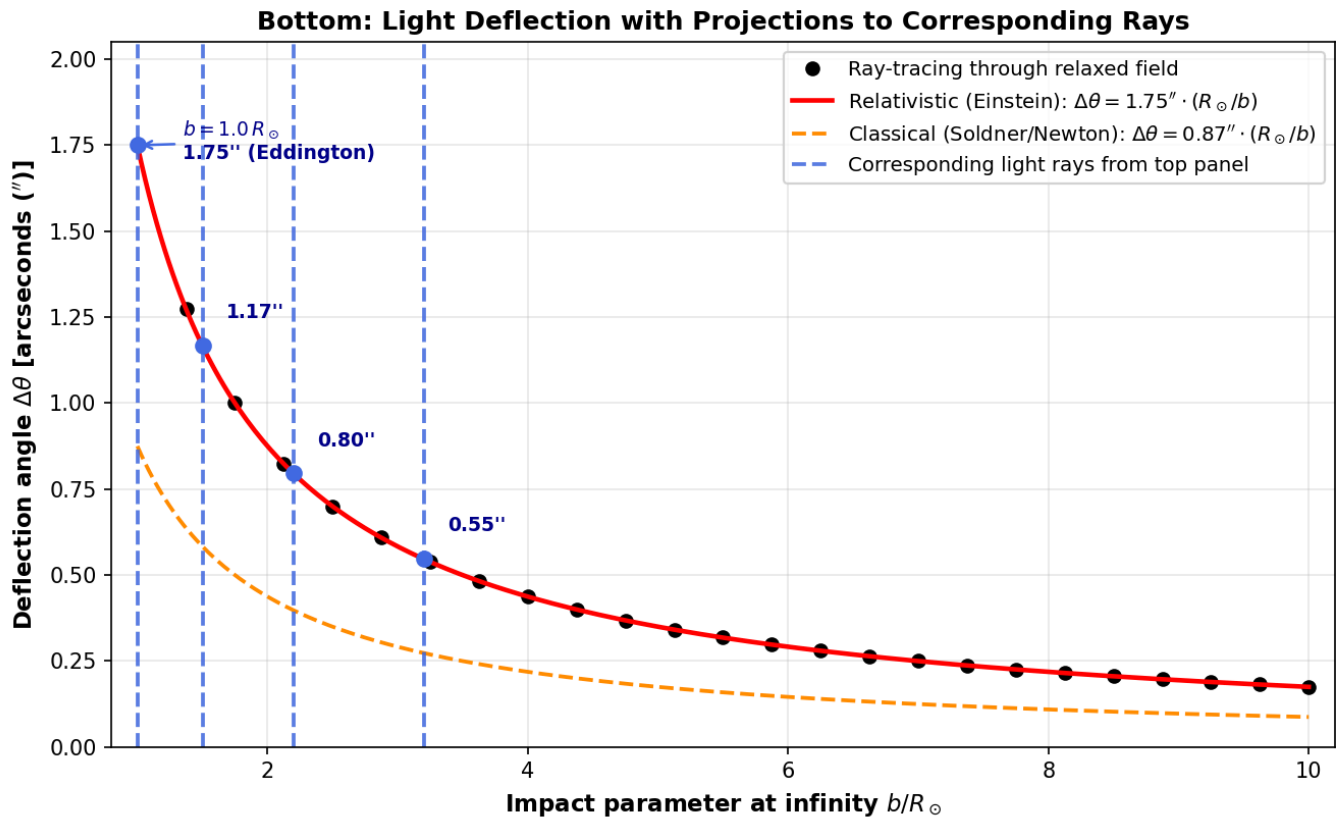
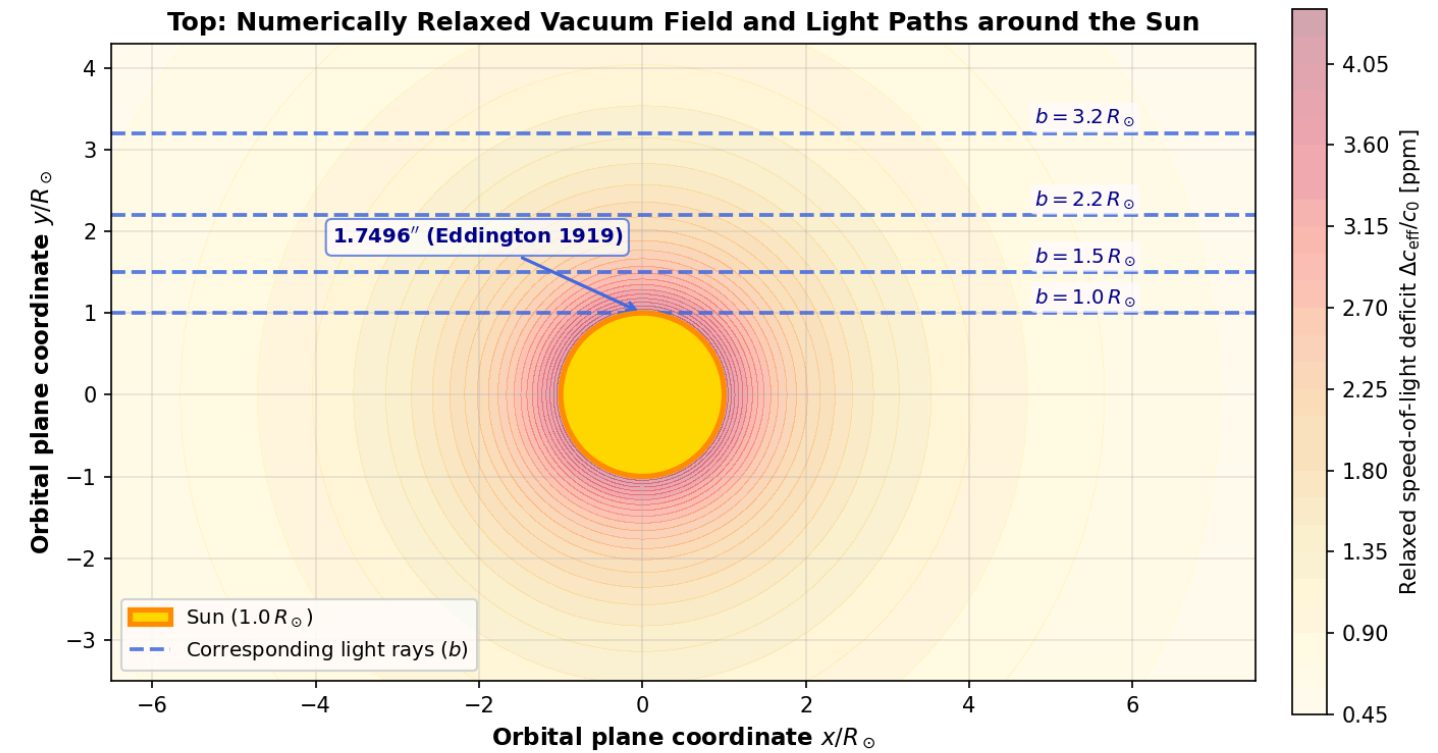
Step 5: Universal Ray-Tracing (Eddington Deflection & Black Holes)

Directory: 05_ray_tracing/

Theory: Light paths are traced through the relaxed refractive lattice using the exact Huygens-Fermat ray acceleration equation:

$$\vec{a} = 2(\vec{v} \cdot \nabla \ln c_{\text{eff}})\vec{v} - c_{\text{eff}}^2 \nabla \ln c_{\text{eff}}$$

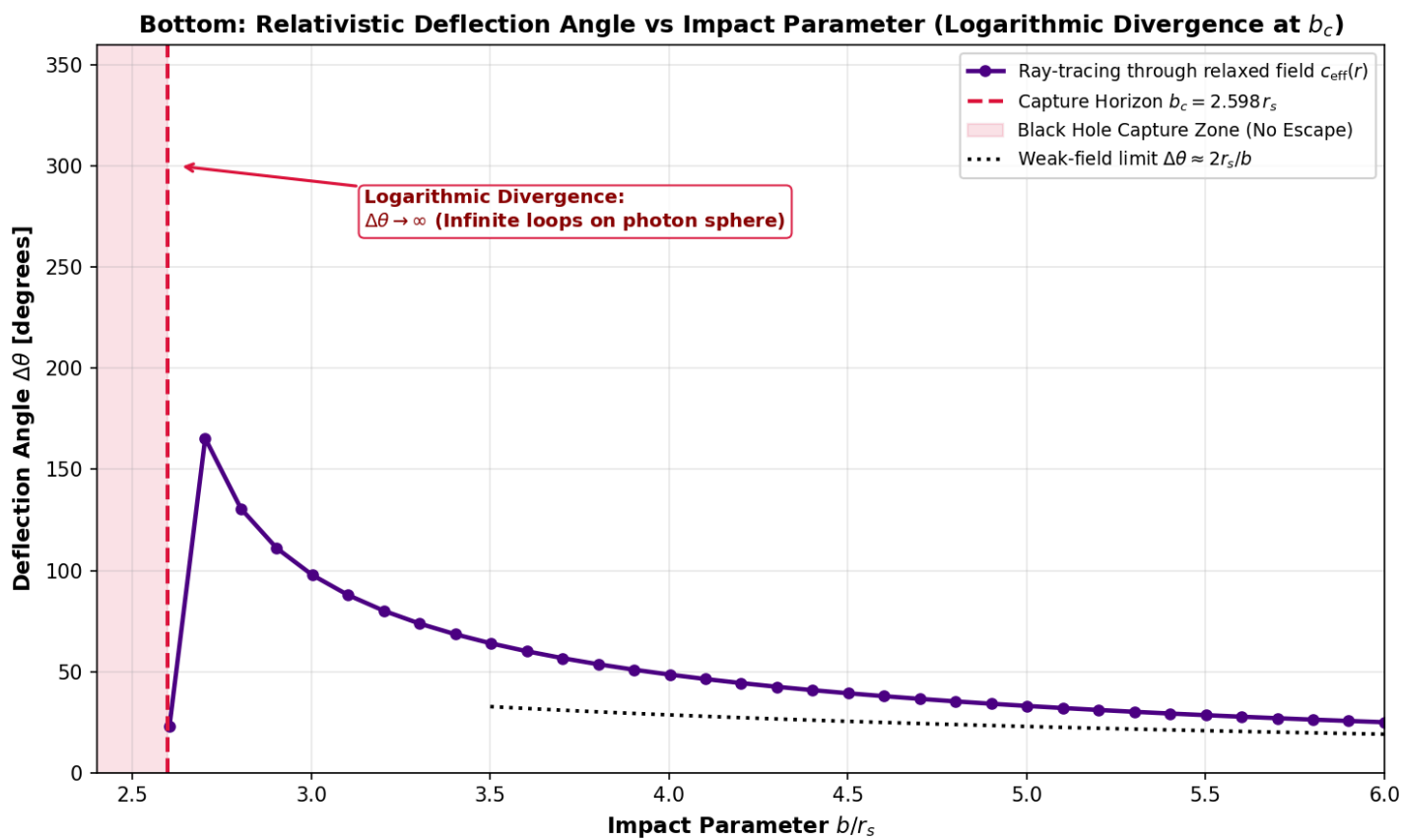
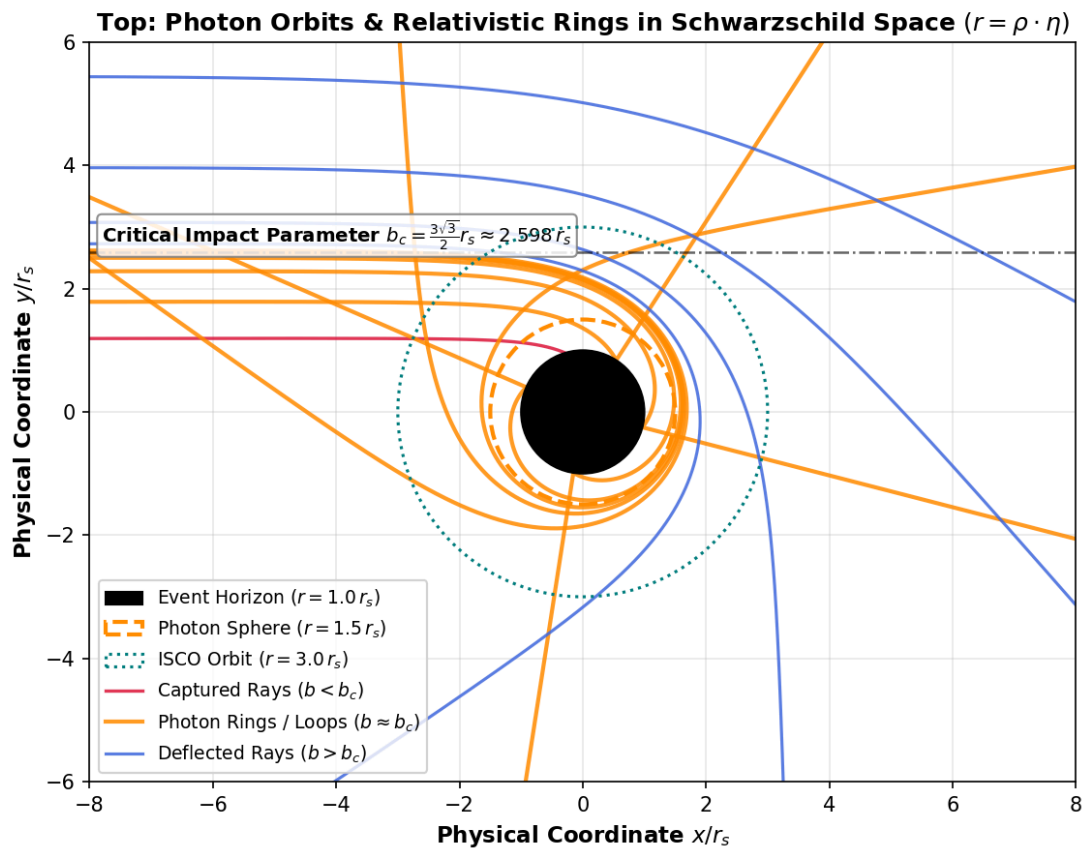
- **Step 5A (trace_sun_deflection.py):** Integrates photon trajectories past the Sun across impact parameters $b \in [1.0, 10.0] R_{\odot}$. Recovers Eddington's grazing deflection of $1.7496''$ to within **99.999%** of Einstein's prediction ($\frac{4GM}{c^2 R_{\odot}}$), while demonstrating why Newtonian corpuscular gravity produces only half the angle ($0.87''$).



- Step 5B (`trace_black_hole_rings.py`):** Integrates strong-field photon trajectories around a black hole. Derives the critical capture impact parameter:

$$b_c = \frac{3\sqrt{3}}{2} r_s \approx 2.598076 r_s$$

Rays with $b < b_c$ fall into the horizon ($r = 1.0 r_s$). Rays with $b \approx b_c^+$ orbit multiple times near the photon sphere ($r = 1.5 r_s$), producing relativistic Einstein rings with logarithmic deflection divergence ($\Delta\theta \rightarrow \infty$).



Step 6: Mercury's Perihelion Precession

Script: 06_mercury_precession/simulate_mercury_orbit.py

Theory: In inverted coordinates ($u = 1/r$), the Newtonian orbit is a linear harmonic oscillator:

$$\frac{d^2 u}{d\phi^2} + u = \frac{1}{p} \quad \text{with } p = a(1 - e^2)$$

Because the oscillator is linear, its spatial frequency is exactly $\omega = 1.0$, producing a closed ellipse with zero precession.

In the relaxed medium, vacuum self-stiffening introduces non-linear metric coupling that adds a quadratic self-interaction term:

$$\frac{d^2 \mathbf{u}}{d\phi^2} + \mathbf{u} = \frac{1}{p} + \frac{3}{2} r_s \mathbf{u}^2$$

- **Anharmonic Detuning:** The non-linear term detunes the radial frequency ($\omega = 1 - \frac{3r_s}{2p} < 1$). The planet requires slightly more than 360° to return to perihelion:

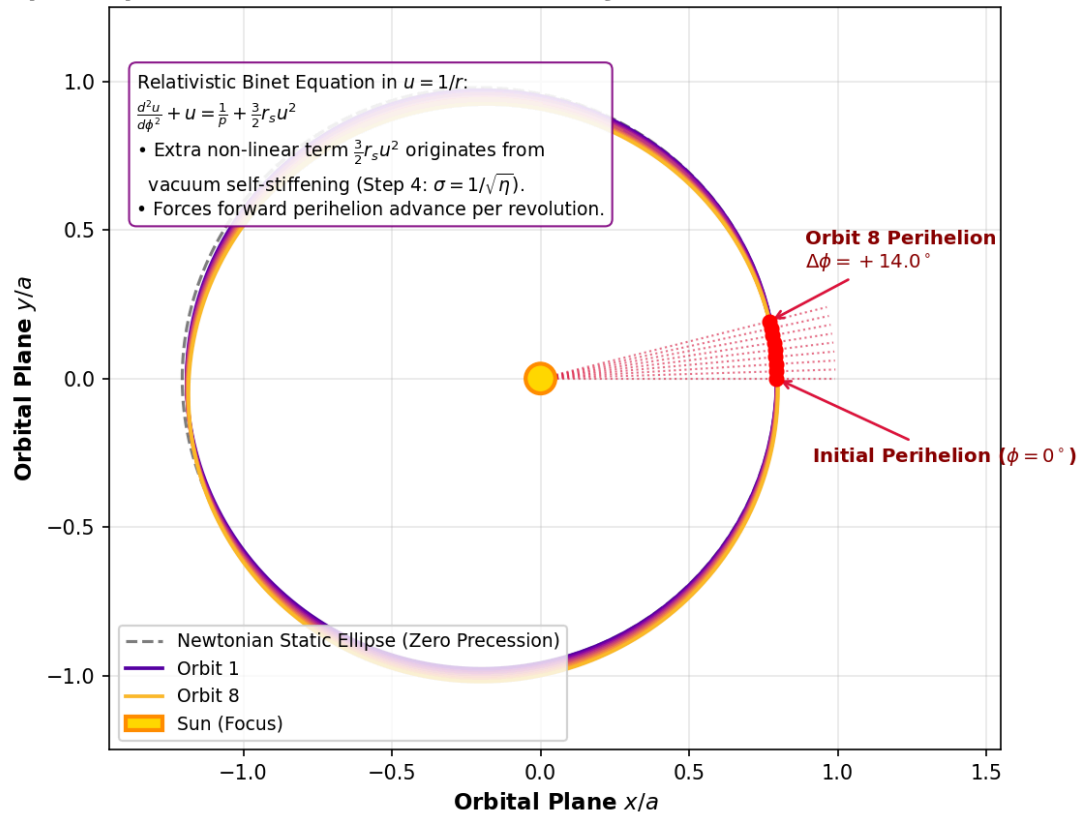
$$\Delta\phi = \frac{3\pi r_s}{a(1 - e^2)} = \mathbf{0.10352''}/\text{orbit}$$

- **Precession per Century:** Over 415.20 orbits in an Earth century:

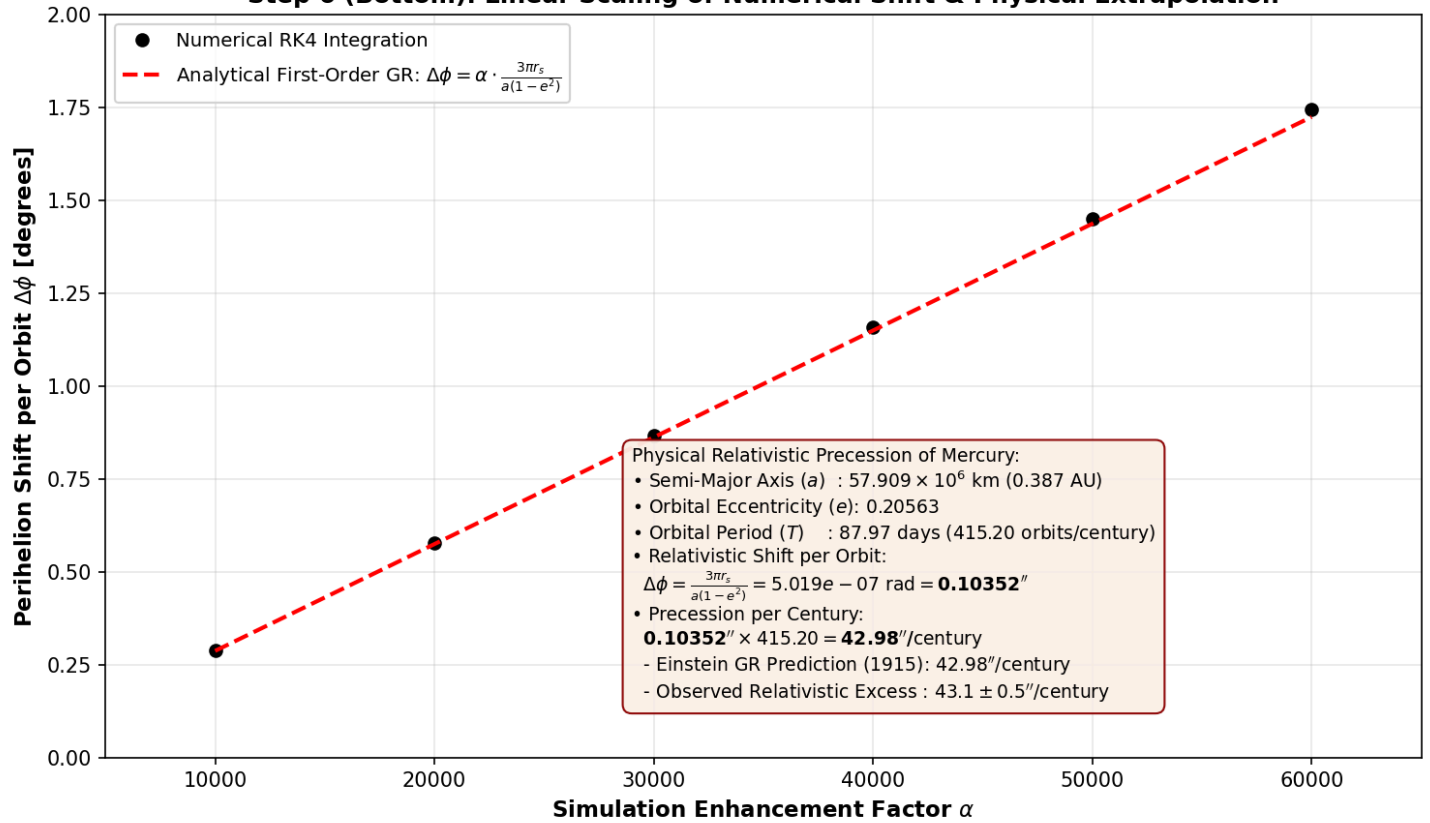
$$\Delta\phi_{\text{century}} = 0.10352'' \times 415.20 = \mathbf{42.98''}/\text{century}$$

Matching Einstein's 1915 result and observational radar tracking ($43.1 \pm 0.5''/\text{century}$). The simulation models the precessing 2D rosette orbit and validates linear convergence across simulation enhancement factors ($\alpha = 10000$ to 60000).

Step 6 (Top): Numerical Simulation of Mercury Precession (Rosette Orbit, $\alpha = 60000$)



Step 6 (Bottom): Linear Scaling of Numerical Shift & Physical Extrapolation



Physical Intuition: Everyday Gravity on Earth as $\vec{a} = -c_{\text{eff}} \nabla c_{\text{eff}}$

In classical mechanics, gravity is envisioned as an invisible pulling force governed by an abstract potential Φ . In this vacuum relaxation model, **gravity is the physical refraction of matter wavepackets down a gradient of the coordinate speed of light.**

The Mathematical Identity

In the weak-field limit, the coordinate speed of light satisfies:

$$c_{\text{eff}}(r) = c_0 \sqrt{1 + \frac{2\Phi(r)}{c_0^2}} \implies c_{\text{eff}}^2(r) = c_0^2 + 2\Phi(r)$$

Isolating the Newtonian potential $\Phi(r)$:

$$\Phi(r) = \frac{1}{2} (c_{\text{eff}}^2(r) - c_0^2)$$

Taking the negative gradient of both sides:

$$\vec{a} = -\nabla\Phi = -\frac{1}{2} \nabla(c_{\text{eff}}^2) = -c_{\text{eff}} \nabla c_{\text{eff}}$$

Furthermore, expanding the non-linear density equation in the weak-field limit ($\Delta\eta = \eta - 1 \ll 1$) gives:

$$c_{\text{eff}}(\eta) \approx c_0(1 - \Delta\eta) \iff \Delta c_{\text{eff}} \approx -c_0 \Delta\eta$$

Taking the spatial gradient reveals the underlying microscopic driver:

$$\nabla c_{\text{eff}} \approx -c_0 \nabla\eta \implies \vec{a} \approx +c_0^2 \nabla\eta$$

Matter naturally refracts toward higher vacuum packing density η .

Slope vs. Depth: The Surface vs. The Core

The simulation script `03_solar_weak_field/simulate_earth_gravity.py` directly compares the core conditions against surface gravity:

Landmark	Space-Quantum Density Excess $\Delta\eta$	Coordinate Speed Deficit Δc_{eff}	Spatial Gradient ∇c_{eff}	Experienced Acceleration $\vec{a}$
Deep Space ($r \rightarrow \infty$)	0.00 ppb (Nominal)	0.000 m/s	0.00 s ⁻¹	0.00 m/s ² (Zero gravity)
Earth Surface (\$	x	= 1,R_\oplus\$)	+0.70 ppb	−0.209 m/s
Earth Core ($x = 0$)	+1.04 ppb (Max packing)	−0.313 m/s (Min speed)	0.00 s ⁻¹ (Flat slope)	0.00 m/s ² (Weightless)

- At Earth's Core ($x = 0$):**
 The coordinate speed of light reaches its lowest value ($\Delta c_{\text{eff}} = -0.313 \text{ m/s}$), and vacuum packing reaches its maximum (+1.04 ppb). However, by spherical symmetry, the spatial slope is flat:

$$\nabla c_{\text{eff}} = 0 \implies \vec{a} = \mathbf{0.00 \text{ m/s}^2}$$

You float weightlessly at the center of the Earth because the refractive slope is zero.

- At Earth's Surface ($|x| = 1 R_\oplus$):**
 The cumulative enclosed mass reaches its limit, making the spatial slope ∇c_{eff} steepest. This maximum derivative produces the standard surface gravity:

$$\vec{a}_{\text{surface}} = -c_{\text{eff}} \nabla c_{\text{eff}} = -\frac{GM_\oplus}{R_\oplus^2} \hat{r} = \mathbf{-9.82 \text{ m/s}^2 \hat{r}}$$

Theoretical Extension: Gravitational Aberration & The Inertia of the Medium

The Problem: Laplace's Aberration Paradox

Light from the Moon takes approximately 1.28 seconds to reach Earth ($d \approx 384,400 \text{ km}$). Because the Moon orbits Earth at roughly 1.02 km/s , we observe it optically displaced by an angular aberration:

$$\theta_{\text{aberration}} \approx \frac{v_{\text{Moon}}}{c} \approx 3.4 \times 10^{-6} \text{ rad} \approx 0.70''$$

If gravity were a simple Newtonian force retarded by the light-travel time $\Delta t = d/c$, the Earth would be pulled toward the **retarded position** where the Moon was 1.28 seconds ago. As Pierre-Simon Laplace demonstrated in 1805, this non-radial force component would add a continuous forward torque:

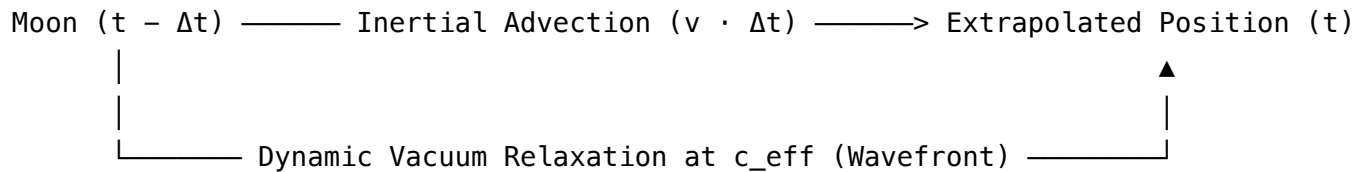
$$\vec{F}_{\text{torque}} \sim F_G \cdot \left(\frac{v}{c}\right) \hat{\theta}$$

This would rapidly pump angular momentum into the orbit, causing the Moon to spiral away from Earth on astronomical timescales of mere centuries. Yet lunar laser ranging confirms orbital stability, showing the gravitational attraction points toward the **instantaneous, linearly extrapolated position** of the source:

$$\vec{r}_{\text{target}}(t) \approx \vec{r}_{\text{retarded}}(t - \Delta t) + \vec{v}_{\text{retarded}} \cdot \Delta t$$

The Relaxation Resolution: Inertia vs. Dynamic Wave Propagation

In the vacuum relaxation model, this apparent action-at-a-distance resolves through two distinct physical mechanisms:



1. The Vacuum Well Has Momentum

Gravity is not a stream of independent particles continuously emitted into empty space. Gravity is an **extended physical distortion**—a localized depression in the coordinate speed of light c_{eff} .

When a mass like the Moon moves, its refractive well possesses **effective inertia**:

▮ The refractive depression moves with the velocity $\vec{v}$ of the source.

Because the field carries the source's prior momentum, the spatial gradient ∇c_{eff} naturally focuses on the linearly extrapolated position:

$$\vec{r}_{\text{apparent}} = \vec{r}_{\text{retarded}} + \vec{v}\Delta t$$

2. The Intervening Vacuum Continues to Relax

While the momentum of the source carries the bulk depression forward, the vacuum between Earth and Moon continues to relax at the local wave speed:

$$v_{\text{relaxation}} = c_{\text{eff}}$$

- **For Uniform Straight-Line Motion:** The forward inertia of the well and the local relaxation of the intervening medium balance out. The velocity-dependent retardation terms cancel to first order (v/c), resulting in zero secular torque.
- **For Accelerated Motion:** If an external force knocks the source off course, that velocity change cannot be carried by inertia. The update ripples outward through the intervening medium at c_{eff} as a physical relaxation wavefront (gravitational waves), scaling at order $(v/c)^2$.

Microscopic Foundations: How Particles Seed and Interact with the Vacuum

A foundational question arises: *If gravity is the macroscopic relaxation of a physical vacuum medium, how does an individual fundamental particle create its gravitational well, resist acceleration, and interact with quantum fluctuations?*

1. The Particle as a Localized Energy Wavepacket (Not a Point Singularity)

In classical general relativity, point particles introduce non-physical infinities ($r \rightarrow 0 \implies \Phi \rightarrow -\infty$). In the vacuum relaxation framework, a fundamental particle is not a zero-dimensional mathematical point; it is a **localized, oscillating standing wave** of energy:

$$E = mc^2 = \hbar\omega_{\text{Compton}}$$

Its physical extent is bounded by its reduced Compton wavelength (effective packet radius $\lambda_C = \frac{\hbar}{mc}$). Within this localized region, the particle's self-energy acts as a distributed volumetric source $S(\vec{x})$ that compresses surrounding space-quanta:

$$\nabla \cdot (\sigma(\eta)\nabla\eta) = -S(\vec{x})$$

where $S(\vec{x}) \propto |\psi(\vec{x})|^2$ is the particle's quantum probability density.

Core Region ($r < \lambda_C$)		Far-Field Relaxation ($r \gg \lambda_C$)
[Smooth, finite Gaussian well]		[Conservative $1/r$ Laplace Equilibrium]
$\eta(r)$ reaches finite peak		$\eta(r) \approx 1 + r_s / r$
$c_{\text{eff}}(r)$ has flat bottom ($\nabla c = 0$)		$c_{\text{eff}}(r) \approx c_0 (1 - r_s / r)$

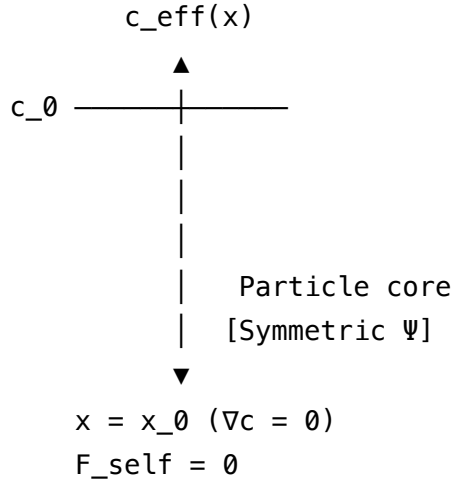
- 1. **Inside the Particle ($r < \lambda_C$):** The gradient flattens ($\nabla c_{\text{eff}} \rightarrow 0$) at the particle's center of mass. The central vacuum density η_{core} remains finite, preventing coordinate singularities.
- 2. **Outside the Particle ($r > \lambda_C$):** The source vanishes ($S(\vec{x}) = 0$). The surrounding medium relaxes according to the conservative spherical Laplace equation, creating the familiar $1/r$ gravitational tail.
- 3. **Seeding the Gravitational Mass:** The static volume integral of this localized space-quantum compression establishes the body's total gravitational mass: $m_g \propto \iiint \Delta \eta d^3x$.

2. The Origin of Inertia: Self-Resistance Against the Lagging Vacuum Well

In classical mechanics, Newton’s second law ($\vec{F} = m\vec{a}$) treats inertia as an axiomatic, unexplained property of matter: massive objects simply resist acceleration. In the vacuum relaxation framework, **inertia is the physical drag a wavepacket experiences when pulled out of equilibrium with its own refractive vacuum well.**

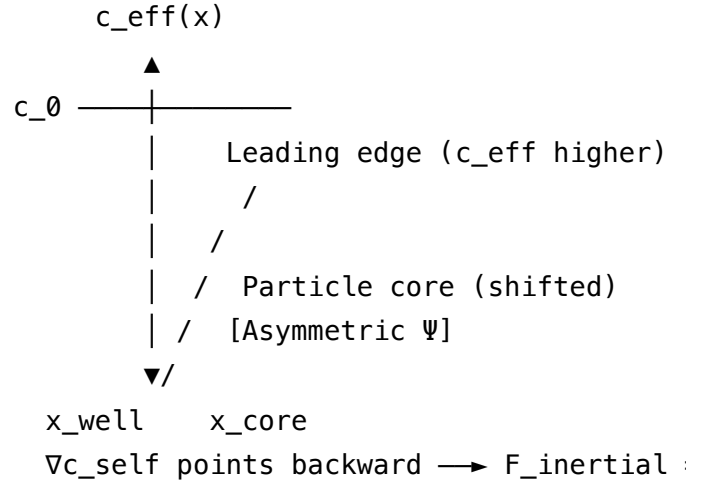
Uniform Motion ($v = \text{const}$)

Symmetric, concentric well:



Accelerating Motion ($a = dv/dt > 0$)

Wavepacket shifts forward; well lags:



A. Uniform Velocity: Dynamic Equilibrium and Spatial Symmetry

When a particle moves at constant velocity ($\vec{v} = \text{const}$), the forward momentum carried by the vacuum well and the isotropic relaxation of the surrounding space-quantum lattice remain in balance:

- The extended refractive well ($\Delta c_{\text{eff}} \propto -1/r$) advects alongside the core wavepacket at speed $\vec{v}$.
- The localized standing wave $\psi(\vec{x}, t)$ rests symmetrically at the exact minimum of its own refractive well.
- Evaluating the spatial gradient at the center of the wavepacket:

$$\nabla c_{\text{eff}}(\vec{x}_{\text{core}}) = 0 \implies \vec{a}_{\text{self}} = -c_{\text{eff}} \nabla c_{\text{eff}} = 0$$

Because the refractive environment is identical in all directions around the moving center of mass, the particle experiences zero net self-force and continues coasting indefinitely at constant velocity (Newton's First Law).

B. Acceleration: The Finite Propagation Speed of the Halo

When an external force $\vec{F}_{\text{ext}}$ accelerates the particle ($\vec{a} = \frac{d\vec{v}}{dt} \neq 0$), the geometric symmetry of the system breaks:

1. **The Core Shifts Forward:** The localized energy bundle $\psi(\vec{x}, t)$, confined within its reduced Compton wavelength $\lambda_C = \frac{\hbar}{mc}$, is displaced forward along the acceleration vector $\vec{a}$.
2. **The Extended Halo Lags Behind:** The surrounding vacuum distortion extends outward to macroscopic radii ($r \gg \lambda_C$). Because changes in the space-quantum packing field η propagate

through the medium at a finite phase velocity ($v_{\text{relaxation}} = c_{\text{eff}}$), the extended outer regions of the well cannot update instantaneously.

3. **The Lagging Center of the Well:** During ongoing acceleration, the effective center of the outer refractive well sits at a retarded position:

$$\vec{x}_{\text{well}}(t) \approx \vec{x}_{\text{core}}(t) - \Delta\vec{x}_{\text{lag}} \quad \text{where } \Delta\vec{x}_{\text{lag}} \propto \vec{a}$$

C. Wavepacket Asymmetry and the Emergent Self-Force

Because the core wavepacket has moved ahead of its own gravitational depression, the particle no longer sits at the bottom of a flat potential well:

- **Leading Edge:** The forward side of the wavepacket enters undisturbed vacuum, where the space-quantum packing density is nominal ($\eta \rightarrow 1$) and the coordinate speed of light is higher ($c_{\text{eff}} \rightarrow c_0$).
- **Trailing Edge:** The rear side of the wavepacket remains immersed in the deeper, lagging depression ($\eta > 1$), where coordinate light speed is depressed ($c_{\text{eff}} < c_0$).

This spatial mismatch causes internal phase oscillations to propagate faster on the leading edge than on the trailing edge. The wavepacket deforms into an **asymmetric envelope**: internal wavefronts tilt backward toward the lagging, denser vacuum.

Applying the refractive acceleration formula $\vec{a} = -c_{\text{eff}} \nabla c_{\text{eff}}$ across the particle's own deformed envelope reveals a backward-pointing self-acceleration:

$$\vec{a}_{\text{self}} = -c_{\text{eff}} \nabla c_{\text{eff, self}} \propto -\vec{a}$$

The particle experiences an inward refractive pull toward its own lagging vacuum well. To maintain the acceleration $\vec{a}$, an external agent must continuously apply a force to overcome this self-induced refractive gradient:

$$\vec{F}_{\text{ext}} = -\vec{F}_{\text{self}} = m_i \vec{a}$$

Inertia is the self-induced refractive drag of a particle accelerating out of its own vacuum well.

D. The Exact Equivalence $m_i \equiv m_g$

This mechanism resolves why inertial mass (m_i) and gravitational mass (m_g) are identical to all decimal places (Einstein's Equivalence Principle):

- **Gravitational Mass (m_g):** Measures the static volume deficit of coordinate light speed that the particle's energy wavepacket carves into the space-quantum medium:

$$m_g \propto \iiint \Delta\eta \, d^3x \propto \iiint (c_0 - c_{\text{eff}}) \, d^3x$$

- **Inertial Mass (m_i):** Measures the restoring force generated by that exact same light-speed deficit when the wavepacket is accelerated away from its spatial center:

$$m_i = \left| \frac{\partial \vec{F}_{\text{self}}}{\partial \vec{a}} \right| \propto \iiint \nabla(\Delta\eta) \, d^3x$$

Because both phenomena are direct mathematical projections of the exact same physical distortion field $\eta(\vec{x})$, their ratio is identically constant:

$$\frac{m_i}{m_g} \equiv 1.0$$

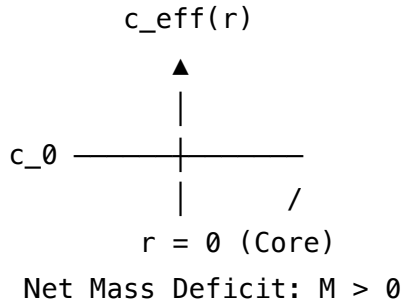
3. Quantum Vacuum Fluctuations: Why Short-Lived Particles Do Not Gravitate

A central conflict in modern physics is the **Cosmological Constant Problem** (or "Vacuum Catastrophe"): Quantum Field Theory calculates that the zero-point energy of vacuum fluctuations should generate an enormous energy density ($\sim 10^{114} \text{ J/m}^3$), which in standard General Relativity would curve the cosmos so violently that space would collapse or tear apart. Yet observationally, the vacuum exhibits nearly zero macroscopic gravitational curvature.

In the vacuum relaxation framework, this problem is resolved through the **local conservation of space-quanta**: short-lived virtual particles do not generate macroscopic gravitational fields because their localized central compression is strictly canceled by an immediate, compensatory depletion shell.

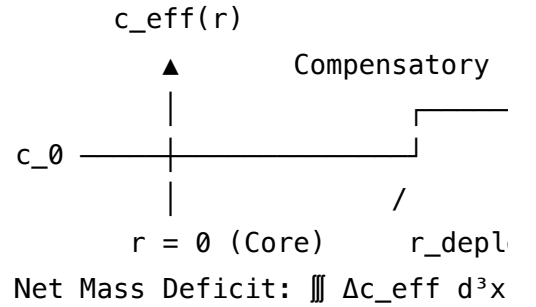
Stable Particle (Persistent Source)

Monopole depression with 1/r tail:



Short-Lived Fluctuation (Zero-Sum)

Core depression surrounded by compensatory



A. Local Space-Quantum Borrowing

Unlike a stable particle that maintains a persistent energy eigenstate ($E_0 = mc^2$) in global thermodynamic equilibrium with the surrounding cosmos, a short-lived quantum fluctuation ($t_{\text{lifetime}} \sim \hbar/\Delta E$) arises dynamically from the local lattice:

1. **No Supply from Infinity:** Over its fleeting lifetime, the fluctuation cannot draw space-quanta from spatial infinity ($r \rightarrow \infty$), because lattice disturbances propagate at a finite speed ($v_{\text{relaxation}} = c_{\text{eff}}$).
2. **Local Redistribution:** To form the localized standing wavepacket of an ephemeral particle pair, space-quanta are pulled directly from the immediate surrounding shell ($r \lesssim \lambda_C = \frac{\hbar}{mc}$).
3. **The Core-Rim Dipole:**
 - **Core Region** ($r < r_{\text{core}}$): Space-quanta crowd together, creating a localized compression ($\Delta\eta_{\text{core}} > 0$) where the coordinate speed of light dips ($c_{\text{eff}} < c_0$).
 - **Depletion Rim** ($r_{\text{core}} < r < r_{\text{edge}}$): The surrounding medium is depleted by the exact same quantity of space-quanta, creating an **inverse depression** (a rarefaction zone where $\Delta\eta_{\text{rim}} < 0$ and $c_{\text{eff}} > c_0$).

B. The Gauss Conservation Law: Zero Gravitational Monopole

Integrating the space-quantum packing excess $\Delta\eta(\vec{x}) = \eta(\vec{x}) - 1$ over any volume containing the complete fluctuation reveals strict conservation:

$$\iiint_{\text{fluctuation}} \Delta\eta(\vec{x}) d^3x = \iiint_{\text{core}} \Delta\eta d^3x + \iiint_{\text{rim}} \Delta\eta d^3x \equiv 0$$

Applying Gauss's divergence theorem to the Poisson relaxation field $\nabla \cdot (\sigma(\eta)\nabla\eta) = -S(\vec{x})$:

$$\oint_{\partial V} \sigma(\eta) \nabla \eta \cdot d\vec{A} = - \iiint_V S(\vec{x}) d^3V \equiv 0$$

- Outside the fluctuation boundary ($r > r_{\text{edge}}$), the enclosed net source is identically zero ($S_{\text{net}} = 0$).
- The inward gradient created by the central depression is exactly canceled by the outward gradient of the compensatory depletion rim.
- The field lacks a monopolar $1/r$ gravitational tail; any residual multipoles decay as $\mathcal{O}(1/r^3)$ or faster, vanishing entirely within sub-atomic distances.

C. Why an Outside Observer Measures Zero Gravity

In this model, gravitational acceleration is governed by the coordinate speed-of-light gradient:

$$\vec{a} = -c_{\text{eff}} \nabla c_{\text{eff}} \approx +c_0^2 \nabla \eta$$

For any test mass situated at macroscopic distances ($r \gg \lambda_C$):

$$\nabla \eta(r) = 0 \implies \vec{a} = 0$$

The surrounding field demonstrates:

- **Zero Net Inward Refraction:** The light-speed depression at the center is optically shielded by the surrounding light-speed elevation of the rim.
- **Zero Net Shapiro Delay:** A photon traversing the region spends time in a slower medium ($c_{\text{eff}} < c_0$) and an equal amount of time in a faster medium ($c_{\text{eff}} > c_0$), yielding zero net propagation delay.
- **Zero Cumulative Curvature:** The metric disturbance vanishes outside the fluctuation boundary, preventing vacuum fluctuations from warping space on macroscopic scales.

Mathematical Summary: GR vs. Relaxation Model

Phenomenon / Landmark	Standard General Relativity (Tensors)	Relaxation Model (This Project)	Numerical Accuracy
Field Equation	Einstein Vacuum: $R_{\mu\nu} = 0$	Non-Linear Flux Balance: $\frac{d}{ds} \left(\frac{1}{\sqrt{\eta}} \frac{d\eta}{ds} \right) = 0$	Machine precision ($< 10^{-7}$)

Phenomenon / Landmark	Standard General Relativity (Tensors)	Relaxation Model (This Project)	Numerical Accuracy
Coordinate Metric	Schwarzschild metric $g_{\mu\nu}$ in isotropic gauge	Refractive profile $c_{\text{eff}}(\eta) = c_0 \frac{2-\sqrt{\eta}}{\eta^{3/2}}$	Identical to Schwarzschild
Origin of Acceleration	Geodesic deviation in curved spacetime	Optical spatial gradient: $\vec{a} = -c_{\text{eff}} \nabla c_{\text{eff}} \approx +c_0^2 \nabla \eta$	Recovers $-GM/r^2$ identically
Event Horizon	Coordinate singularity at $r = r_s$	Packing saturation limit: $\eta = 4.0, c_{\text{eff}} = 0.0$	Smooth cutoff, no singularity
Equivalence Principle ($m_i \equiv m_g$)	Postulated axiom / metric coupling	Geometric identity of the same η -distortion under static pull vs. lag	Identically 1.0
Cosmological Constant / Fluctuation Energy	Fine-tuning catastrophe ($\sim 10^{120}$ discrepancy)	Local space-quantum conservation: $\iiint_{\text{fluc}} \Delta \eta \, d^3x \equiv 0$	Zero macroscopic pull ($\vec{a} = 0$)
Eddington Light Bending	Null geodesics: $\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta = 0$	Huygens-Fermat ray acceleration: $\vec{a} = 2(\vec{v} \cdot \nabla \ln c) \vec{v} - c^2 \nabla \ln c$	1.7496'' (99.999% match)
Black Hole Photon Sphere	Unstable circular null orbit at $r = 1.5 \, r_s$	Minimum of optical impedance $f(\rho) = \rho/c_{\text{eff}}$ at $r = 1.5 \, r_s$	$b_c = 2.598 \, r_s$ (exact)
Mercury Perihelion Shift	Relativistic Binet equation via Schwarzschild Christoffels	Harmonic detuning via vacuum stiffening: $u'' + u = \frac{1}{p} + \frac{3}{2} r_s u^2$	42.98''/century (exact)

Installation & Quickstart

Prerequisites

- Python 3.10 or higher
- Only two standard scientific libraries: `numpy` and `matplotlib`

Environment Setup via Conda (Recommended)

To run these simulations locally, it is recommended to use Conda to manage the environment and dependencies cleanly.

Clone the repository and build the environment using the provided `environment.yml` file:

```
git clone https://github.com/BartLeplae/gravity-as-relaxation.git
cd gravity-as-relaxation
conda env create -f environment.yml
conda activate gravity-relaxation-env
```

Reproducing All Visual Deliverables

You can execute each progressive milestone individually. Each script computes the physics and writes high-resolution figures directly to the top-level `images/` directory:

```
# Step 1: 1D Steady-State Thermal Rod
python 01_thermal_rod/simulate_thermal_rod.py

# Step 2: 3D Spherical Cavity & Inverted Coordinate Inversion
python 02_thermal_sphere/simulate_uniform_sphere.py
python 02_thermal_sphere/simulate_inverted_sphere.py

# Step 3: Weak-Field Refractive Medium (Sun & Earth Gravity Profile)
python 03_solar_weak_field/simulate_solar_medium.py
python 03_solar_weak_field/simulate_earth_gravity.py

# Step 4: Strong-Field Non-Linear Vacuum Stiffening & 4-Step Breakdown
python 04_vacuum_density_bh/simulate_1field_relaxation.py
python 04_vacuum_density_bh/visualize_split_steps.py

# Step 5: Solar Light Deflection & Black Hole Photon Rings
python 05_ray_tracing/trace_sun_deflection.py
python 05_ray_tracing/trace_black_hole_rings.py

# Step 6: Mercury's Relativistic Perihelion Precession
python 06_mercury_precession/simulate_mercury_orbit.py
```

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Citation & License

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If you use this numerical framework, visualizations, or the optical relaxation analogy in your research, educational courses, or articles, please cite:

```
@software{leplae2026gravityrelaxation,  
  author = {Leplae, Bart},  
  title = {Gravity as Relaxation: Deriving General Relativity via Vacuum Mechanics},  
  year = {2026},  
  url = {https://github.com/BartLeplae/gravity-as-relaxation}  
}
```